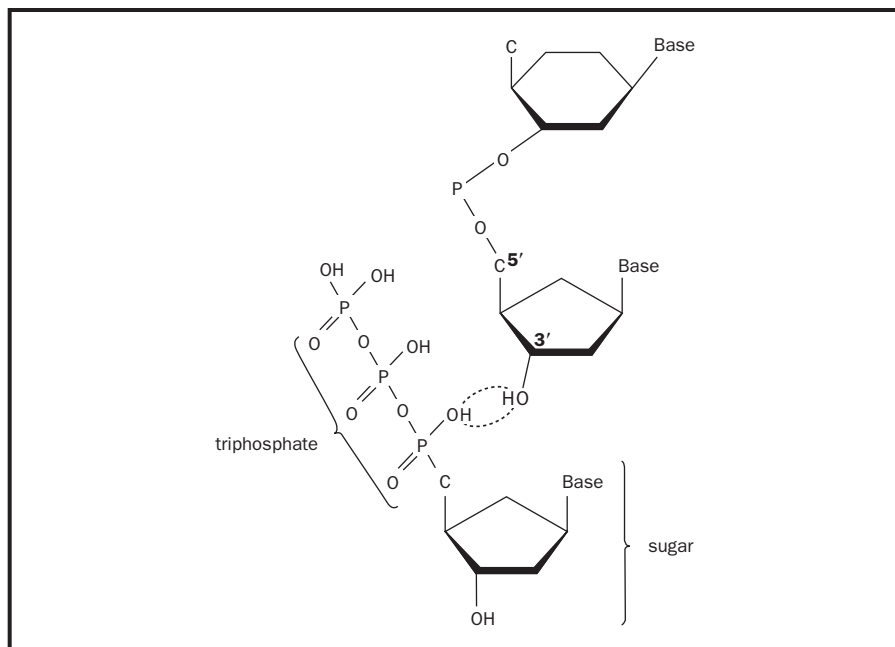




DNA polymerase attaches an incoming triphosphate nucleotide to the 3' position on the growing chain. Removal of a water molecule between the two attaches them, and the two terminal phosphates break off.

above diagram). Therefore, DNA polymerization occurs in only one direction. Some DNA polymerases are highly processive, polymerizing many nucleotides to the 3' end of the DNA chain before falling off the DNA template. Other DNA polymerases are distributive in nature, incorporating just one nucleotide and then falling off the DNA template.

Occasionally, the DNA polymerase will incorrectly polymerize a nucleotide onto the growing DNA chain. Removal of this misinserted nucleotide must be performed by a “proofreading” **exonuclease**, which is a substance that removes nucleotides from the 3' end of the DNA molecule. The combined actions of DNA polymerases and proofreading exonucleases improve the accuracy of DNA synthesis and thus minimize introduction of errors into the genome.

exonuclease enzyme that cuts DNA or RNA at the end of a strand

The Variety of DNA Polymerases

Like all proteins, the DNA polymerases are encoded in genes. The genes that encode the human DNA polymerases are contained in the genomic DNA at various positions on several different chromosomes. In February 2001 the first “working draft” sequence of the human genome was published. Analysis of this sequence shows us that there are perhaps as many as fifteen different DNA polymerase genes in the human genome. Each of these genes encodes a different DNA polymerase protein. However, biochemists have not yet isolated all of these enzymes. A similar analysis of the *Escherichia coli* genome shows us that there are five different DNA polymerase genes present in this bacterium. The multitude of DNA polymerases in human and bacterial cells indicates a specialized role for the different enzymes in various aspects of DNA replication and repair, many of which have yet to be identified.

The human DNA polymerase α is encoded in the *POLA* (polymerase alpha) gene located on the human X chromosome. The DNA polymerases δ and ϵ are encoded in the *POLD1* (polymerase delta 1) and *POLE1*

mitochondria energy-producing cell organelle

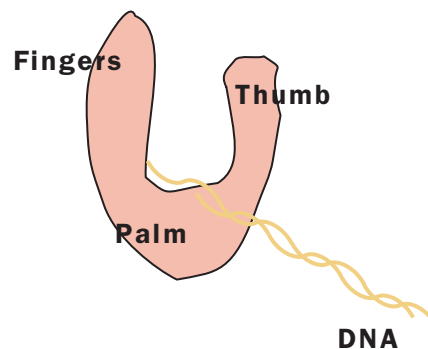
catalytic describes a substance that speeds a reaction without being consumed

(polymerase epsilon 1) genes, which are located on chromosomes 19 and 12, respectively. These three DNA polymerases are most frequently associated with replication of the human genome. The DNA polymerase β is encoded by the *POLB* (polymerase beta) gene on chromosome 8 and is involved in DNA repair.

The DNA polymerase γ is encoded by the *POLG* (polymerase gamma) gene on chromosome 15 and replicates the DNA of the **mitochondria**. In *Escherichia coli* the DNA polymerase I is the most active. This enzyme functions in the bacterial cell to repair DNA, while the DNA polymerase III is responsible for replicating the genome. There are several additional DNA polymerases in human and in bacterial cells of which the precise function is not known. Some of these enzymes might be necessary to replicate genomic DNA that has been damaged. The ability to replicate damaged DNA could lead to mutations introduced into the genome but would preserve the life of the cell.

The amino acid sequences of the DNA polymerase proteins can be deduced from the genetic code contained in the DNA polymerase genes. Based on the amino acid sequences of the DNA polymerases, these proteins have been classified into several families. Analysis of these sequences reveals a relatively diverse collection of proteins with some very important similarities in specific amino acid regions along the length of the protein.

The similarities in amino acid sequences in certain parts of the DNA polymerase proteins tell us that these regions of the protein have been conserved throughout evolution. These specific amino acids are those that are important in the **catalytic** function of DNA polymerization by these proteins. Some of the similar amino acids are necessary for binding metal atoms that are needed by the DNA polymerase to carry out the polymerization reaction. Other amino acid sequences allow the DNA polymerase to hold on to the DNA and the four different deoxynucleoside triphosphates as the enzyme polymerizes the new DNA chain. Some DNA polymerases have the necessary amino acid sequences to generate a 3' proofreading exonuclease domain (region), allowing the DNA polymerase to remove mistakes, or proofread, as it builds the DNA polymer.



The DNA polymerases range in size from just over three hundred amino acids in length to more than two thousand amino acids in length. Three-dimensional studies of these enzymes have shown that the DNA polymerases have a common protein fold that resembles the shape of a “right hand” (see diagram). The “thumb,” “fingers,” and “palm” form a pocket along which the DNA can move. The DNA molecule interacts with specific amino acids

located in the “palm” region of the DNA polymerase, and the “thumb” clamps down on the DNA to hold it as the DNA chain-elongation reaction proceeds. The amino acids that are common in many of the DNA polymerases are found in the regions where the enzyme contacts the DNA molecule. SEE ALSO DNA REPAIR; *ESCHERICHIA COLI* (*E. COLI* BACTERIUM); HUMAN GENOME PROJECT; NUCLEASES; NUCLEOTIDE; MUTATION; REPLICATION; X CHROMOSOME.

Fred W. Perrino

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DNA Profiling

DNA profiling is a molecular testing method used to uniquely identify people and other organisms. In many ways, it is similar to blood typing and fingerprinting, and it is sometimes called “DNA fingerprinting.” Because every organism’s DNA is unique, DNA can be examined to identify people who might be related to each other, to compare suspected criminals to DNA left at the scene of a crime, or even to identify certain strains of disease-causing bacteria.

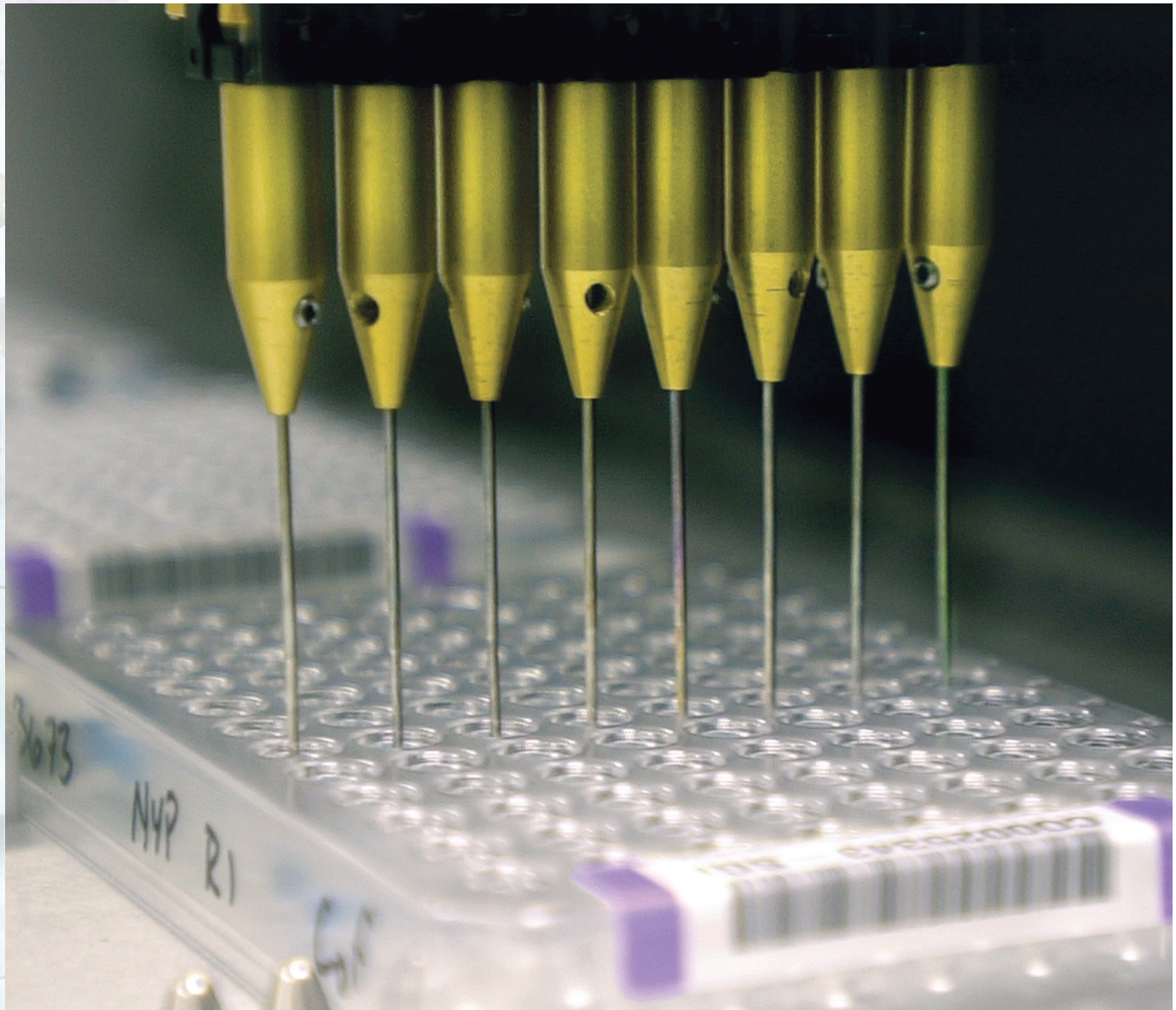
Blood Typing and the ABO Groupings

Before the development of the molecular biology tools that make DNA testing possible, investigators identified people through blood typing. This method hails from 1900, when Karl Landsteiner first discovered that people inherited different blood types. Several decades later, researchers determined that the basis for those blood types was a set of proteins on the surface of red blood cells.

The main proteins on the surface of red blood cells used in blood typing come in two varieties: A and B. Every person inherits from their parents either the genes for the A protein, the B protein, both, or neither. Someone who inherits the A gene from one parent and neither gene from the other parent has blood type A. If a person inherits both genes, they are AB. A person who inherits neither is type O. Another protein group found on red blood cells is referred to collectively as the Rh factor. People either have the Rh factor or they do not, regardless of which of the A and B genes they inherited. To type a person’s blood, **antibodies** against these various proteins (A, B, and Rh) are mixed with a blood sample. If the proteins are present, the blood cells will stick together and the sample will get cloudy.

antibodies immune-system proteins that bind to foreign molecules





A robot used in DNA profiling adds solution and stirs DNA samples from tissues taken from September 11, 2001, New York terrorism victims. The tissue is being identified by matching DNA samples, which is the essence of DNA profiling.

Blood typing can be used to exclude the possibility that a blood sample came from a particular person, if the person's type does not match that of the sample. However, it cannot be used to claim that any particular person is the source of the sample, because there are so few blood types, and they are shared by so many people. About 45 percent of people in the United States are type O, and another 40 percent are type A. If four people were physically present at the scene of a murder, and the candlestick found nearby had type O blood spilled on it, chances are good that two of those individuals could be found guilty of the crime, based solely on the blood typing evidence. Most court cases, however, rely on more evidence than just blood or DNA typing, such as whose fingerprints are also found on the candlestick (see Statistics and the Prosecutor's Fallacy, below).

DNA Polymorphism Offers High Resolution

DNA is the molecule that contains all the genetic information of an individual. One person's DNA is made up of about three billion building blocks

known as nucleotides or bases. Every organism in the world has a unique DNA sequence except for identical twins. Although identical twins accrue changes as they develop, they generally do not accumulate enough genetic differences for DNA typing to be useful. Portions of the DNA, called genes, encode proteins within the sequence of bases. Genes are separated by long stretches of noncoding DNA. Because these sequences do not have to code for functional proteins, they are free to accumulate more differences over time, and thus provide more variation than genes. Thus, they are more useful than gene sequences in distinguishing individuals.

Polymorphisms are differences between individuals that occur in DNA sequences which occupy the same **locus** in the chromosome. An individual will have only one sequence at a particular polymorphic locus in each chromosome, but if the population bears several to dozens of different possible sequences at the site in question, then the locus is considered “highly variable” within the population. DNA profiling determines which polymorphisms a person has at a small number of these highly variable loci. Because of this, DNA profiling can provide high resolution in distinguishing different individuals. The chances of one person having the same DNA profile as another are typically much less than the chances of winning a lottery.

STR Analysis

The technology of DNA profiling has advanced from its beginnings in the 1980s. Today, DNA profiling primarily examines “short tandem repeats,” or STRs. STRs are repetitive DNA elements between two and six bases long that are repeated in tandem, like GATAGATAGATAGATA. These repeat sequences often exist in a chromosomal region called heterochromatin, a largely unused portion of DNA found in each chromosome.

Different STR sequences (also called genetic markers) occur at different loci. While their positions are fixed, the number of repeated units varies within the population, from four to forty depending on the STR. Therefore, one genetic marker may have between four and forty different variations, and each variation is referred to as an allele of that marker. Each person has at most two **alleles** of each marker, one inherited from each parent. The two alleles for a particular marker may be identical, if both parents had the same form.

The United States Federal Bureau of Investigation has designated thirteen of these sequences to use with STR analysis. These thirteen markers are all four-base repeats, and were chosen because multiple alleles of each exist throughout the population. The FBI system, called CODIS (Combined DNA Indexing System), has become the standard DNA profiling system in use today.

STR analysis begins with sample collection. Because of the often small samples involved and the legal weight that will be given to them, it is vital that the sample not be contaminated by other DNA. This may occur for instance if skin cells from the person collecting the sample are mixed with skin cells under the fingernails of a victim. Once the sample is collected, it must be kept secure at all times, to prevent any possibility of tampering.

In the laboratory, the DNA is isolated and purified, and then multiple copies of it are made using the polymerase chain reaction (PCR).

polymorphisms occurring in several forms

locus site on a chromosome (plural, loci)

alleles particular forms of genes

PCR polymerase chain reaction, used to amplify DNA



Technicians can specify which DNA sequences to multiply, so that only the thirteen core STR sequences will be amplified (multiple copies produced), leaving the rest of the billions of irrelevant bases alone.

In order to specify which DNA to amplify, “primers” are used. The primers are DNA sequences that recognize a nonrepeated sequence in the genetic markers, and which are used by the DNA polymerase that does the actual copying. After the DNA has been copied, the new DNA molecules are separated by size, by gel electrophoresis. A fluorescent molecule previously attached to each primer will send a light signal to the machine that measures the length of the molecule, or allele.

VNTR Analysis

An early form of DNA profiling, rarely used today, is based on VNTRs, or “variable number of tandem repeats.” VNTRs requires extensive sample processing: The DNA is chopped up with restriction enzymes, separated by size, and probes are applied to the fragmented DNA to view only the relevant DNA pieces. In the DNA of two different individuals, different spacing between two cut sites for the restriction enzymes gives a unique pattern of DNA size fragments, called “restriction fragment length polymorphisms,” or RFLPs.

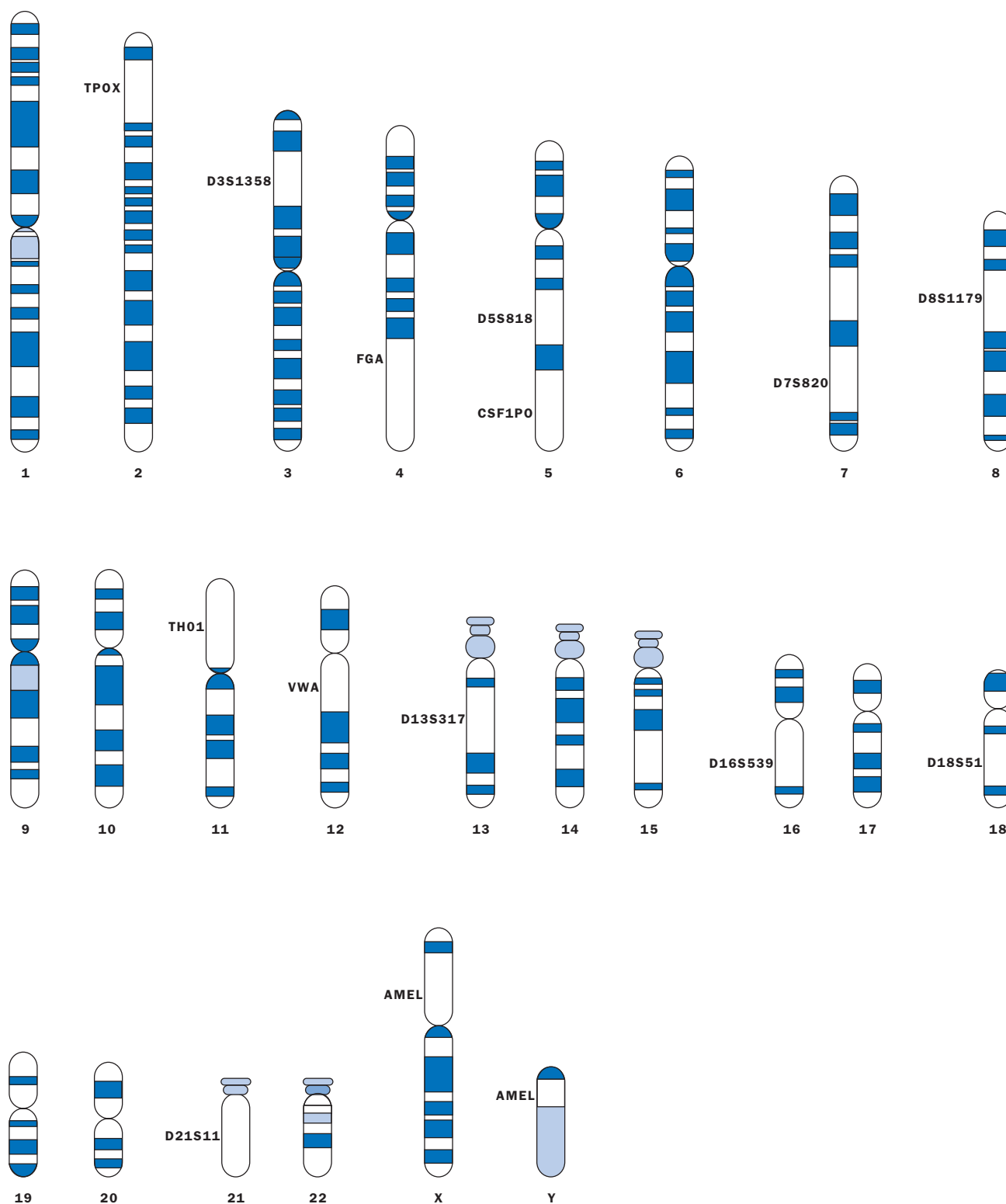
Making a Match

To understand how DNA profiling is used to identify a person, imagine a sample of blood collected at a crime scene that doesn’t match the victim’s blood, and is presumably from the unknown perpetrator. DNA from the blood is isolated and its set of STRs are analyzed. The results will be a list of the alleles found at each of the markers (for example, VWA–12, 13; TH01–6, 7, and so on), where the initial symbol is the abbreviation for the markers and the last two are the numbers of the alleles found in the sample for that marker. The full set of thirteen markers may or may not be analyzed in each case. When a suspect is identified, his or her DNA can be analyzed for these same markers. If the set of alleles are different, the investigators can be sure that the two DNAs came from different sources, and the suspect is not the source of the blood. Since the introduction of DNA profiling, an absence of matching DNA has been used to free dozens of wrongly convicted prisoners.

If the samples do match, the question becomes whether the blood is actually from the suspect, or from someone else with the same set of alleles. As with blood typing, this is a matter of statistics, and depends on how frequently each allele occurs in the population. This information has been tabulated and is kept on file in the FBI CODIS database. If two samples share a very rare allele, that increases the likelihood they came from the same source.

Matching multiple alleles increases the certainty they came from the same source. Since the thirteen STRs are inherited independently of each other, the likelihood that one person’s DNA will include specific alleles of all thirteen STR sites is the product of the individual allele frequencies. For example, if each allele a person carries occurs in 25 percent of the population, then the probability that all thirteen alleles will occur in one

13 CORE CODIS STR LOCI WITH CHROMOSOMAL POSITIONS



Locations of the FBI's 13 core loci used for DNA profiling. The loci were chosen to span the genome, and to have maximum variability among humans. Adapted from <<http://www.cstl.nist.gov/biotech/strbase/images/codis.jpg>>.

Statistics and the Prosecutor's Fallacy

Let's assume the DNA profile found at the crime scene—and the matching DNA of the suspect—is expected to occur once in every million people. The correct statement of probability arising from these facts is, “If the suspect is innocent, there is a one-in-one-million chance of obtaining this DNA match.” The fallacy is to reverse these clauses, and state, “If the DNA matches, there is a one in one million chance that the suspect is innocent.” To understand the logical fallacy, imagine the statement, “If it's Tuesday, it must be a school day.” The reverse is not true—there are other school days besides Tuesday.

DNA Profiling Comes of Age

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DNA testing can also be used in organisms other than humans. For instance, it has been used to type cattle in a cattle-stealing case. It can also be used to identify pathogenic strains of bacteria to track the outbreak of disease epidemics. SEE ALSO BIOTECHNOLOGY AND GENETIC ENGINEERING, HISTORY; GEL ELECTROPHORESIS; POLYMERASE CHAIN REACTION; POLYMORPHISMS; REPETITIVE DNA ELEMENTS.

Mary Beckman

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DNA Repair

When it was discovered that DNA is the **macromolecular** carrier of essentially all genetic information, it was assumed that DNA is extremely stable. Consequently, it came as something of a surprise to learn that DNA is actually unstable and subject to continual damage. When DNA damage is severe, the cell is unable to replicate and may die. Repair of DNA must be regarded as essential for the preservation and transmission of genetic information in all life forms. In this article, we will discuss various types of DNA damage and the DNA repair systems that have evolved to correct that damage.

Sources of Damage

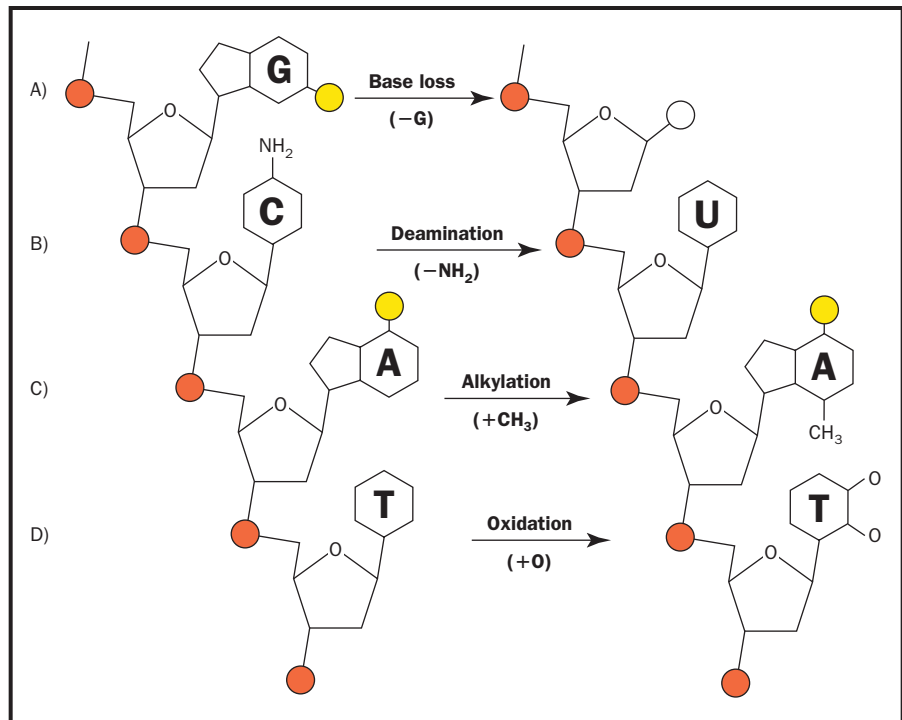
DNA is subject to spontaneous instability and decay. In addition to spontaneous damage, cellular DNA is under constant attack from reactive chemicals that the cell itself generates as by-products of **metabolism**. Moreover, the integrity of cellular DNA is assaulted by such environmental threats as X rays, ultraviolet radiation from the sun, and many chemical agents, some of which are products of our industrialized society.

Since mutations can be introduced into DNA as a consequence of DNA damage, there is currently great interest and concern about the expanding list of chemicals released into the environment. In humans, damage to DNA has been implicated in many cancers as well as in certain aspects of aging. Genetic diseases such as cystic fibrosis and sickle cell disease can be caused by a single DNA mutation in one gene.

macromolecular large molecule, composed of many similar parts

metabolism chemical reactions within a cell

Figure 1. Four types of mutation: (A) complete loss of a base. (B) Loss of an amino group, converting a cytosine to a thymine. (C) Addition of a small alkyl group, such as $-\text{CH}_3$. (D) Reaction with oxygen.



Types of DNA Damage

Damage to DNA can result from several different types of processes. Hydrolysis, deamination, alkylation, and oxidation are all capable of causing a modification in one or more bases in a DNA sequence.

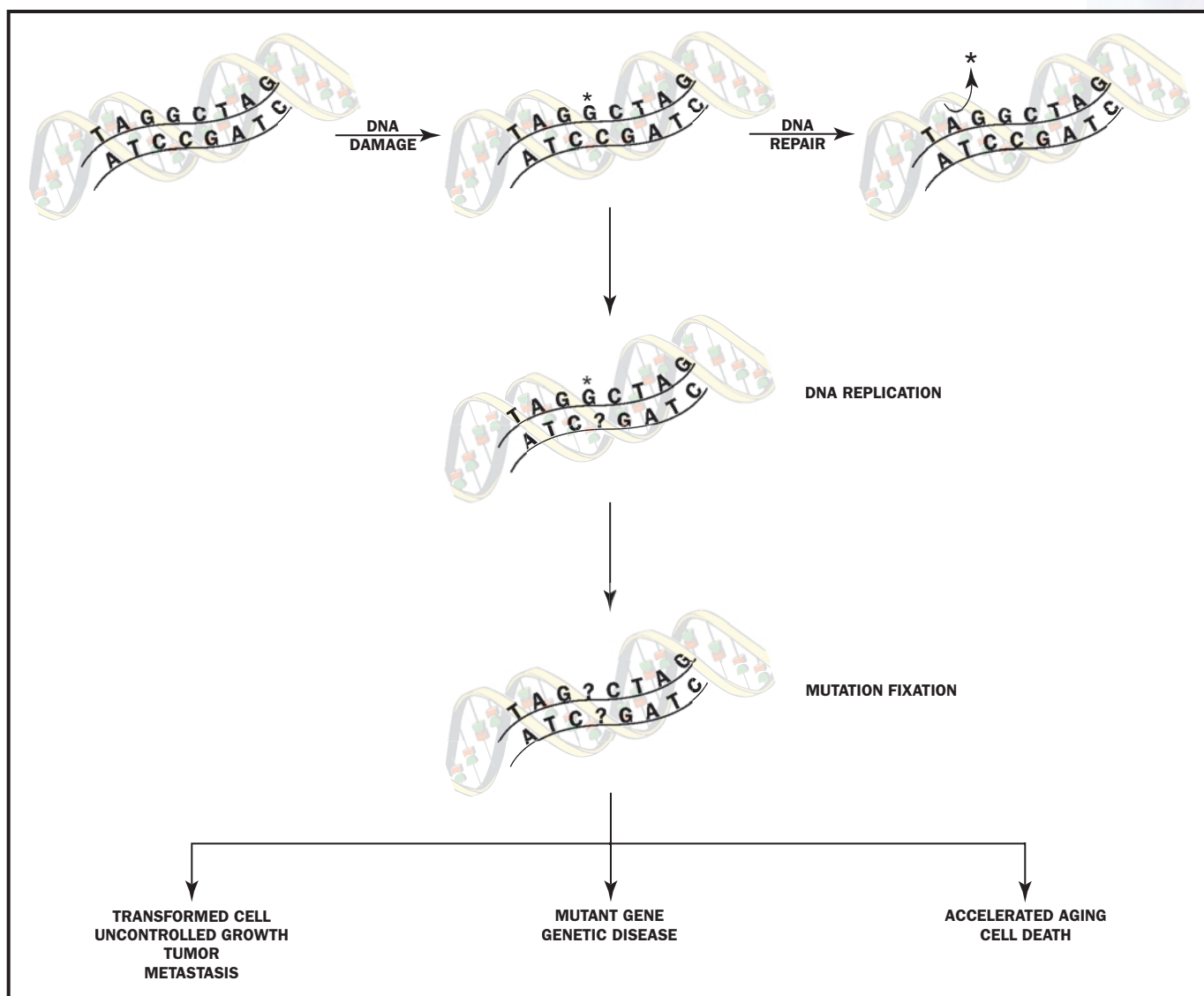
Hydrolysis. DNA consists of long strands of sugar molecules called deoxyribose that are linked together by phosphate groups. Each sugar molecule carries one of the four natural DNA bases: adenine, guanine, cytosine, or thymine (A, G, C, or T). The chemical bond between a DNA base and its respective deoxyribose, although relatively stable, is nonetheless subject to chance **cleavage** by a water molecule in a process known as spontaneous **hydrolysis**. Loss of the “purine” bases (guanine and adenine) is referred to as depurination, whereas loss of the “pyrimidine” bases (cytosine and thymine) is called depyrimidination. In mammalian cells, it is estimated that depurination occurs at the rate of about 10,000 purine bases lost per cell generation. The rate of depyrimidination is considerably slower, resulting in the loss of about 500 pyrimidine bases per cell generation.

The baseless sugars that result from these processes are commonly referred to as AP-sites (apurinic/aprimidinic). They are potentially lethal to the cell, as they act to block the progress of DNA replication, but are efficiently repaired in a series of enzyme-catalyzed reactions collectively referred to as the base excision repair (BER) pathway. In fact, AP-sites are intentionally created during the course of BER.

Deamination. The bases that make up DNA are also vulnerable to modification of their chemical structure. One form of modification, called spontaneous deamination, is the loss of an amino group ($-\text{NH}_2$). For example, cytosine (C), which is paired with guanine (G) in normal, double-stranded DNA, has an amino group attached to the fourth carbon (C4) of the base.

cleavage hydrolysis

hydrolysis splitting with water



When that amino group is lost, either through spontaneous, chemical, or enzymatic hydrolysis, a uracil (U) base is formed, and a normal C–G DNA base pair is changed to a premutagenic U–G base pair (uracil is not a normal part of DNA).

The U–G base pair is called premutagenic because if it is not repaired before DNA replication, a mutation will result. During DNA replication, the DNA strands separate, and each strand is copied by a DNA **polymerase** protein complex. On one strand, the uracil (U) will pair with a new adenine (A), while on the other strand the guanine (G) will pair with a new cytosine (C). Thus, one DNA double-strand contains a normal C–G base pair, but the other double-strand has a mutant U–A base pair. This process is called mutation fixation, and the mutation of the G to an A is said to be fixed (meaning “fixed in place,” not “repaired”). In other words, the cell now accepts the new mutant base pair as normal. It is estimated that approximately 400 cytosine deamination events per genome occur every day. Clearly, it is very important for the cell to repair DNA damage before DNA replication commences, in order to avoid mutation fixation. One cause of

Figure 2. If damaged DNA (*) is not repaired, the mistake (mutation) can be replicated and become permanent (fixed) in the genome. This may cause severe problems.

polymerase enzyme complex that synthesizes DNA or RNA from individual nucleotides

mutagen any substance or agent capable of causing a change in the structure of DNA

hormone molecule released by one cell to influence another

endogenous caused by factors inside the organism

excision removal

catalyze aid in the reaction of

normal human aging is the gradual accumulation over time of mutations in our cellular DNA.

Alkylation. Another type of base modification is alkylation (Figure 2C). Alkylation occurs when a reactive **mutagen** transfers an alkyl group (typically a small hydrocarbon side chain such as a methyl or ethyl group, denoted as $-\text{CH}_3$ and $-\text{C}_2\text{H}_5$, respectively) to a DNA base. The nitrogen atoms of the purine bases (N_3 of adenine and N_7 of guanine) and the oxygen atom of guanine (O_6) are particularly susceptible to alkylation in the form of methylation. Methylation of DNA bases can occur through the action of exogenous (environmental) and endogenous (intracellular) agents. For example, exogenous chemicals such as dimethylsulfate, used in many industrial processes and formed during the combustion of sulfur-containing fossil and N-methyl-N-nitrosoamine, a component of tobacco smoke, are powerful alkylating agents. These chemicals are known to greatly elevate mutation rates in cultured cells and cause cancer in rodents.

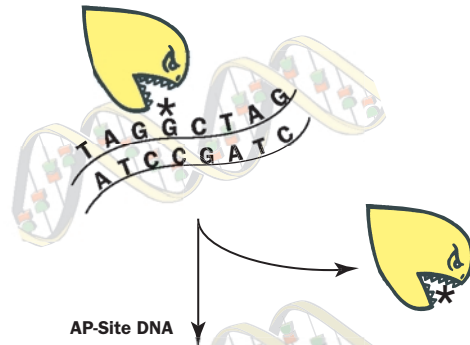
Inside every cell is a small molecule known as S-adenosylmethionine or “SAM.” SAM, which is required for normal cellular metabolism, is an endogenous methyl donor. The function of SAM is to provide an activated methyl group for virtually every normal biological methylation reaction. SAM helps to make important molecules such as adrenaline, a **hormone** secreted in times of stress; creatine, which provides energy for muscle contraction; and phosphatidylcholine, an important component of cell membranes. However, SAM can also methylate inappropriate targets, such as adenine and guanine. Such endogenous DNA-alkylation damage must be continually repaired; otherwise, mutation fixation can occur.

Oxidation. Oxidative damage to DNA bases occurs when an oxygen atom binds to a carbon atom in the DNA base (Figure 2D). High-energy radiation, like X rays and gamma radiation, causes exogenous oxidative DNA base damage by interacting with water molecules to create highly reactive oxygen species, which then attack DNA bases at susceptible carbon atoms. Oxidative base damage is also endogenously produced by reactive oxygen species released during normal respiration in mitochondria, the cell’s “energy factories.”

Humans enjoy a long life span; thus, it would seem that healthy, DNA repair-proficient cells could correct most of the naturally occurring **endogenous** DNA damage. Unfortunately, when levels of endogenous DNA damage are high, which might occur as the result of an inactivating mutation in a DNA repair gene, or when we are exposed to harmful exogenous agents like radiation or dangerous chemicals, the cell’s DNA repair systems become overwhelmed. Lack of DNA repair results in a high mutation rate, which in turn may lead to cell death, cancer, and other diseases. Also, if the level of DNA repair activity declines with age, then the mutational burden of the cell will increase as we grow older.

Base Excision Repair

DNA bases that have been modified by the addition or loss of a small chemical group as described above are repaired by the BER pathway (Figure 3). The BER pathway begins with the **excision** of a damaged base by an enzyme called DNA glycosylase (Figure 3, step 1). DNA glycosylases bind to chemically altered (damaged) bases and **catalyze** the cleavage (hydrolysis) of the

BASE EXCISION REPAIR PATHWAY**1) DNA Glycosylase**

AP-Site DNA

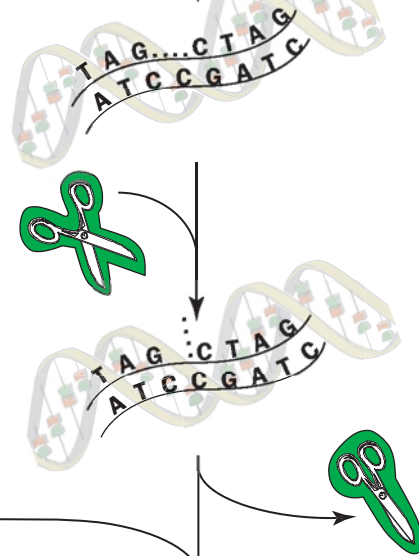
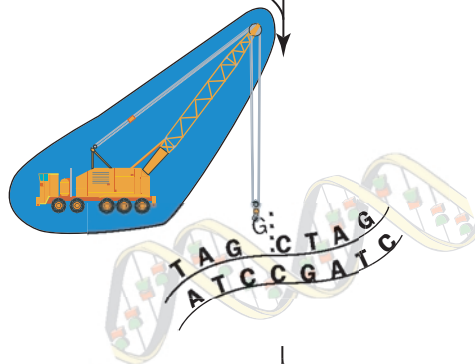
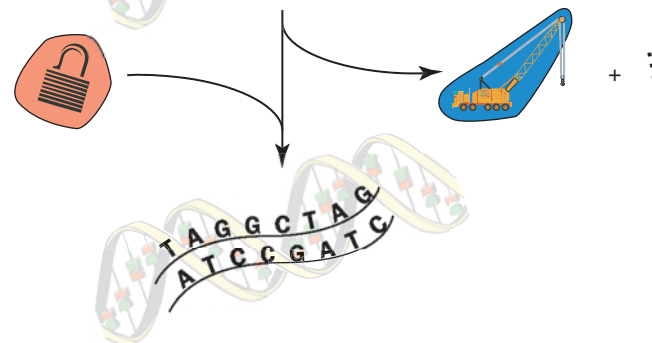
2) AP-Endonuclease**3) DNA Polymerase****4) DNA Ligase**

Figure 3. Schematic diagram of the base excision repair pathway. The damaged base (in this case, guanine) is removed by a glycosylase enzyme. This creates an AP-site, which is recognized and cleaved by AP-endonuclease. DNA polymerase fills in the gap. DNA ligase links the broken strand together again.

bond linking the modified base to its sugar, which results in the release of the modified base from the DNA chain and in the insertion of an AP-site. Several types of DNA glycosylases exist, each one specifically excising a different type of damaged base. It is important that a DNA glycosylase act only on damaged and not natural DNA bases, otherwise too many baseless sugars would be produced, weakening the integrity of the DNA chain.

Excision of the damaged base by a DNA glycosylase creates an AP-site, which in turn is acted upon by the second enzyme in the BER pathway, apurinic/apyrimidinic (AP) endonuclease (Figure 3, step 2). The most abundant AP-endonuclease in human cells cleaves (incises) the sugar-phosphate backbone on the left side of the baseless sugar to yield a one-nucleotide gap. On the left margin of the incision is a normal nucleotide (DNA base + sugar + phosphate); however, the right margin of the gap contains the baseless sugar-phosphate residue.

In order to fill the gap (replace the missing nucleotide), an enzyme specialized in synthesizing DNA, a DNA polymerase, will insert the correct nucleotide into the gap and link it to the normal nucleotide on the left margin by recognizing which base is opposite the gap on the complementary DNA strand. Figure 3, step 3 shows that the DNA polymerase recognizes that a G nucleotide is needed since the complementary base is a C. Note that an entire nucleotide is added here, not just a base. Before DNA polymerase is finished with the repair of the one-nucleotide gap, it removes the baseless sugar phosphate left behind by AP-endonuclease.

At this point, repair of the gap is almost, but not quite, finished, since there is a “nick” in the top DNA strand at the right margin of the former gap. Thus, the final step in the BER pathway is to **ligate** the DNA strands on both sides of the nick (Figure 3, step 4). If we examine the sugar phosphate DNA chain shown in Figure 2, we can see that the sugars that carry the DNA bases are linked together by phosphate groups. This type of linkage is referred to as a **phosphodiester bond**. The enzyme DNA ligase joins the strands by creating a phosphodiester bond between them, sealing the nick. In summary, the basic steps of the BER pathway are damage recognition and base excision, AP-site incision, DNA repair synthesis, and DNA ligation.

ligate join together

phosphodiester bond
the link between two
nucleotides in DNA or
RNA

Nucleotide Excision Repair

DNA damage that involves particularly “bulky” molecules or chemical bonds between bases, or that significantly distorts the double-stranded structure of DNA, is subject to repair by the nucleotide excision repair (NER) pathway. For example, it has long been known that the ultraviolet (UV) light in sunshine can damage DNA by forming what are called photoproducts. UV radiation excites many types of molecules, causing them to react with each other and with DNA. In particular, UV light can catalyze the formation of chemical bonds between adjacent thymine and/or cytosine bases; these bonds are called intra-strand UV crosslinks (Figure 4A). These crosslinked bases distort the double-stranded structure of DNA and block DNA replication.

A second example of bulky DNA damage is that caused by large, organic molecules like aflatoxin, found in mold-contaminated peanuts, and benzo[a]pyrene (Figure 4B), a main component of smoke and soot. Both

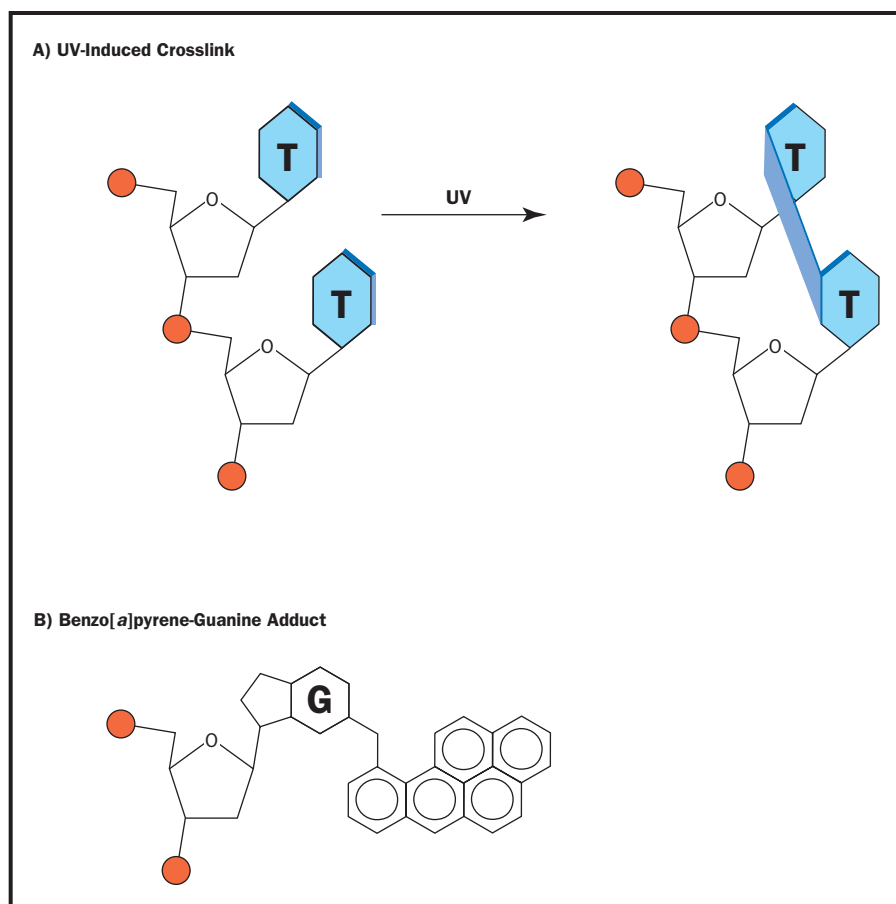


Figure 4. Two types of mutations repaired by the nucleotide excision repair pathway. Ultraviolet light can trigger a chemical reaction that links adjacent thymines. Complex organic molecules such as benzo[a]pyrene can link on to a base such as guanine.

aflatoxin and benzo[a]pyrene are potent **carcinogens**. Ingestion or inhalation of these and similar compounds activates the body's detoxification systems, which convert the **hydrophobic** organic molecules into water-soluble forms for removal. However, the intermediate forms of aflatoxin and benzo[a]pyrene produced during the detoxification reaction happen to be very reactive with DNA purines, and form DNA base adducts (they “add on” to DNA). Specifically, such compounds tend to adduct guanine and, to a lesser extent, adenine. These large DNA adducts can cause mutations, and, since they block DNA replication, deletions of large segments of DNA can occur. Also, they activate the cell's damage surveillance systems, and, if not repaired, can cause cell death (apoptosis).

The mechanism of NER, involving some thirty proteins, is more complex than that of BER, but the basic principles are similar: damage recognition, damage excision, DNA repair synthesis, and DNA ligation (Figure 5). Damage recognition is obviously very important (Figure 5, step 1), but how can a single multiprotein complex detect so many different types of DNA damage? The answer is that the DNA damage must (1) distort the normal double-stranded structure of DNA, and/or (2) block transcription by RNA polymerase. Unusual kinks or twists in double-stranded DNA are recognized by the NER damage-recognition multiprotein complex. Also, when RNA polymerase stalls at a damaged DNA base, components of the NER damage-recognition complex are recruited to the site of damage.

carcinogens substances that cause cancer

hydrophobic “water hating,” such as oils

DNA crosslinks also interfere with another vital cellular process: transcription of genes by RNA polymerase.

ligase enzyme that repairs breaks in DNA

microsatellites small repetitive DNA elements dispersed throughout the genome

Next, the double-stranded DNA adjacent to the damage is unwound by a DNA unwinding enzyme called a helicase (Figure 5, step 2). Unwinding of the DNA allows repair proteins access to the site of damage. The DNA strand containing the damaged base is then cleaved a few nucleotides after the damage, and about twenty-five nucleotides before it, by specific endonucleases associated with the NER protein complex (Figure 5, step 3). Endonucleases are enzymes that cleave inside a segment of DNA.

Next, the DNA segment that contains damage is displaced by DNA polymerase and associated proteins, and a corresponding repair patch is synthesized (Figure 5, step 4). Lastly, DNA **ligase** seals the nick, joining the newly synthesized piece of DNA to the preexisting strand (Figure 5, step 5).

DNA Mismatch Repair

The DNA mismatch repair (MMR) pathway has evolved to correct errors made by DNA polymerase during DNA replication. Such errors fall into two broad categories: base substitutions and insertions/deletions. A base substitution error occurs when DNA polymerase inserts an incorrect (non-complementary) nucleotide opposite the template base, like a T opposite G instead of C, or A opposite C instead of G. These incorrect base pairs are referred to as mispairs or mismatches. Often, DNA polymerase will make a base substitution error when copying a base that has been damaged by alkylation. For example, DNA polymerase will frequently insert a T opposite O⁶-methylguanine on the other strand.

An insertion error occurs when DNA polymerase adds one or more extra nucleotides (+1, +2, +3, and so on) to a sequence; a deletion error is made when one or more nucleotides (−1, −2, −3, and so on) are omitted from a sequence. Sequences that contain repeats of the same nucleotide (mononucleotide repeat), such as AAAAAAAAAA, are particularly vulnerable to +1 or −1 insertion/deletion errors when copied by DNA polymerase. Such sequences might be called “slippery,” in that DNA polymerase can “slide” on the DNA and lose its place. Other repetitive sequences, like the dinucleotide repeat CACACACA and the trinucleotide repeat CTGCTGCTG, are prone to +2 and +3, or −2 and −3 insertion/deletion errors, respectively. These repetitive DNA sequences are called **microsatellites**.

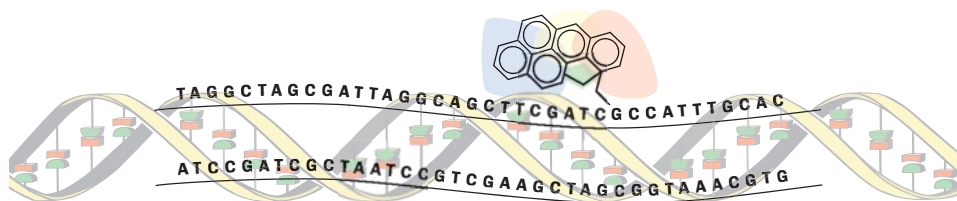
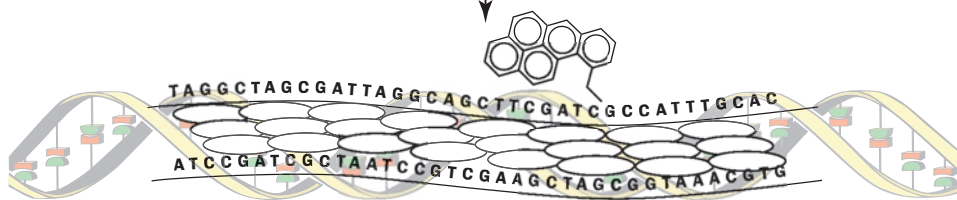
Defects in DNA mismatch repair have been found in several types of cancer, notably colon cancer, and microsatellite sequences that are either shorter or longer than normal are a hallmark of defective MMR. Expansion of trinucleotide repeat sequences is associated with a number of hereditary neurological disorders, such as fragile X syndrome, myotonic dystrophy, and Huntington’s disease.

The process of MMR, like the BER and NER pathways, comprises damage recognition, damage excision, DNA repair synthesis, and DNA ligation. First, a mismatch or insertion/deletion error must be recognized by a complex of proteins specialized for the particular type of damage (mismatch, or small or large insertion/deletion). Just how the mismatch recognition protein complex “knows” which DNA strand contains the “right” nucleotide and/or which DNA strand contains the “wrong” one has not yet been determined.

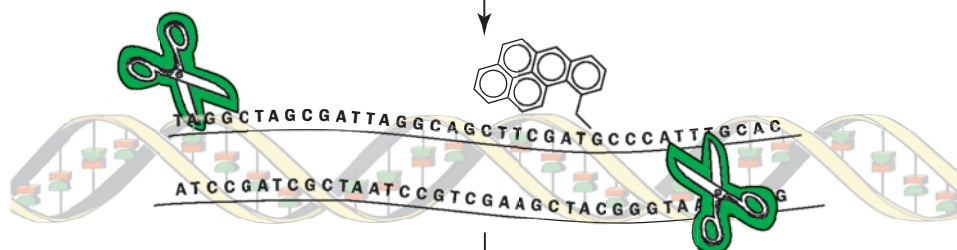
Next, a phosphodiester bond in the DNA strand containing the mismatched nucleotide is cleaved by an endonuclease, the strand is displaced

NUCLEOTIDE EXCISION REPAIR PATHWAY

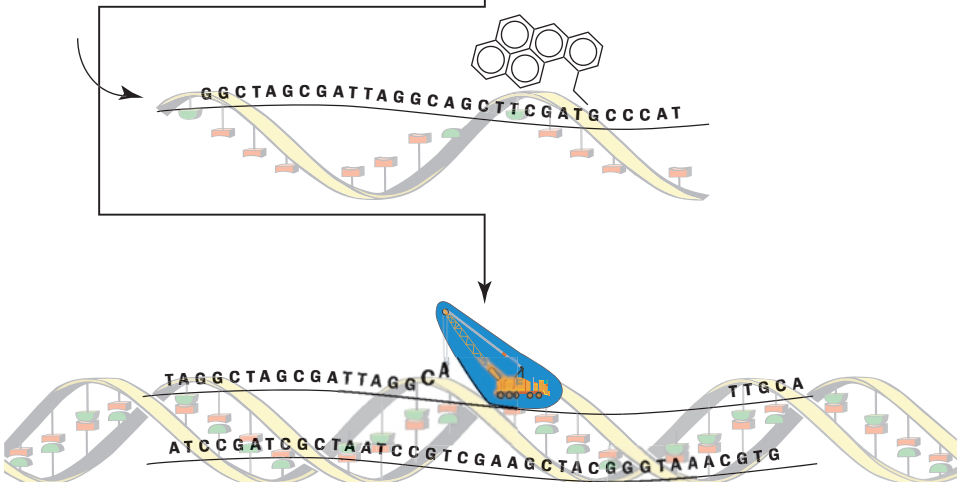
1) Damage Recognition

2) Localized Unwinding
(stabilized by proteins)

3) Nucleotide Excision



3) DNA Polymerase



4) DNA Ligase

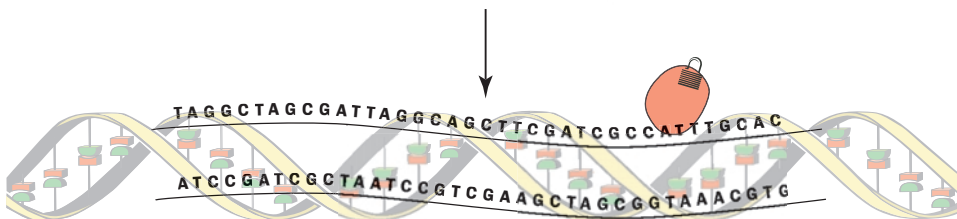


Figure 5. The nucleotide excision repair pathway is used to repair damage that distorts the double helix or prevents replication. After recognition, helicase unwinds the two strands, which are stabilized by single-strand binding proteins. Multiple nucleotides are removed from either side of the damage. DNA polymerase fills the gap, and ligase relinks the strand.

by DNA helicase, and a portion of the strand is removed by a combination of DNA exonuclease and DNA polymerase. Lastly, DNA polymerase carries out DNA repair synthesis, and DNA ligase restores the continuity of the sugar-phosphate-DNA backbone. The patch of DNA newly synthesized by the MMR DNA polymerase is relatively large, approximately 1,000 nucleotides long, compared to the DNA repair synthesis that takes place during BER, which typically replaces 1 nucleotide, or NER, which replaces approximately 30 nucleotides. MMR is especially important in tissues that are constantly regenerating, like the intestinal lining and the endometrium (the lining of the uterus), since growth requires DNA replication, which sometimes makes mistakes.

Future Directions

In addition to the three critical DNA repair pathways already discussed (BER, NER, and MMR), there are two additional types of DNA repair: double-strand break repair and recombinational repair. These are both complex phenomena, and scientists' understanding of them is still at an early stage. Also, many questions about BER, NER, and MMR still await answers. For example, since DNA damage that escapes repair leads to deleterious alterations of our DNA, could we prevent mutation by increasing the levels of DNA repair proteins? Could we live longer and healthier lives with more or better DNA repair? How are DNA repair pathways regulated by the cell? Is there such a thing as too much DNA repair? If repairs always took place whenever DNA damage occurred, would there be no evolution? Exactly how do the proteins and enzymes involved in DNA repair accomplish their jobs? These and many other exciting lines of inquiry are in store for future investigators. SEE ALSO APOPTOSIS; CANCER; CARCINOGENS; DNA POLYMERASES; FRAGILE X SYNDROME; MUTAGEN; MUTATION; NUCLEASES; NUCLEOTIDE; RNA POLYMERASES; TRIPLET REPEAT DISEASE.

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DNA Structure and Function, History

DNA was discovered in the nineteenth century, but its significance as the physical basis of inheritance was not understood until midway through the twentieth. The realization that it was the molecule of heredity led to intensive efforts to determine its three-dimensional structure, and to understand how it stores and transmits genetic information. The discovery of the struc-

ture of DNA, and the elucidation of its function, ranks as one of the greatest achievements of science.

Discovery of DNA

Deoxyribonucleic acid (DNA) was first discovered in 1869 by Johann Friedrich Miescher (1844–1895), a young Swiss chemist studying in Tübingen, Germany. Miescher's interest in the biochemistry of the cell nucleus led him to collect used surgical bandages, from which he collected pus (white blood cells), which have very large nuclei. From these, he purified a new compound, which he termed “nuclein.” Miescher showed that nuclein was a large molecule, acidic, and rich in phosphorus. Miescher continued to work with nuclein over the next two decades, turning for his source to salmon sperm, which have exceptionally large nuclei and were plentiful in the rivers near his laboratory. One of his students renamed the compound “nucleic acid.”

In 1885 the German biologist Oskar Hertwig (1849–1922) suggested that nucleic acid might be the hereditary material, based on its presence in the nucleus and the growing certainty that the nucleus was the center of heredity. Despite this promising beginning, no further progress was made in understanding its true role until the 1940s.

The biochemistry of nucleic acid continued to be studied, however, and by 1900, scientists had learned that it was composed of three parts: a sugar, a phosphate, and a base (together termed a **nucleotide**). The five-carbon sugar is a ringed structure, and it forms an alternating chain with phosphate (a phosphorus atom surrounded by four oxygens). Also attached to the sugar is one of five different bases: adenine, cytosine, guanine, thymine, and uracil, usually abbreviated A, C, G, T, and U. The names of the bases are related to their historical origin: Guanine was isolated from bird guano, thymine from the thymus gland of calves, and adenine from calf pancreas (“adeno-” is a Greek root for gland); uracil is chemically related to urea; and “cyto-” means cell.

In the 1920s it was discovered that there were two types of sugars, ribose and deoxyribose, differing by the presence or absence of one oxygen atom. DNA was shown to incorporate the bases A, C, G, and T, while RNA (ribonucleic acid) incorporates A, C, G, and U. Much of this work was carried out by the German biochemist Albrecht Kossel (1853–1927) and the American Phoebus Aaron Levene (1869–1940).

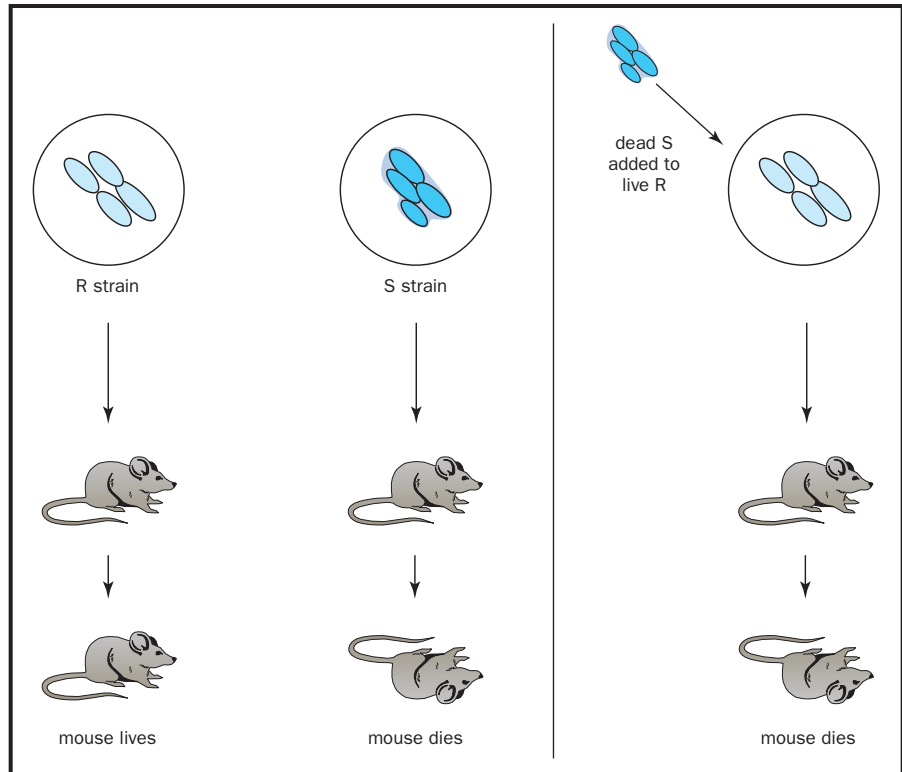
Also by the 1920s, Thomas Hunt Morgan and his colleagues had shown that genes, whatever they were made of, were carried on chromosomes. Chromosomes were shown to contain both protein and DNA, so the question of which of these two substances composed the genes took center stage. The more complex chemical nature of proteins gave them the theoretical edge, an opinion given great and, in hindsight, unfortunate weight by Levene, a widely respected biochemist.

Levene proposed the tetranucleotide hypothesis, which held that the structure of DNA was a monotonous repetition of the four nucleotides in succession. Levene's evidence was that DNA, which he had isolated from a variety of sources, had roughly equal amounts of A, C, G, and T. While the actual proportions he found were not exact, Levene attributed the differences to experimental error, rather than biochemical reality. Such a simple

nucleotide a building block of RNA or DNA



Oswald Avery showed that DNA from dead S strain of *Pneumococcus* bacteria could transform a harmless strain into a deadly strain.



and highly regular molecule as the one Levene proposed could not account for the diversity of life, however, and so DNA was assumed to play only a structural role in the chromosome.

DNA Is the Transforming Factor

DNA was not again taken seriously as the hereditary material until 1944, when Oswald Avery (1877–1955) published a landmark paper outlining his experiments with two strains of *Pneumococcus* bacteria. The S (“smooth”) strain was able to cause disease in mice, while the R (“rough”) strain was not. Under the microscope, the S strain had a smooth, glistening surface, due to a sugar capsule it secreted. The R strain lacked the capsule.

Fifteen years earlier, Frederick Griffith had shown that injecting R bacteria plus heat-killed S bacteria into mice would cause disease just as surely as injecting live S bacteria, a result that Griffith attributed to a “transforming principle.” When Avery grew R bacteria in a dish with the heat-killed S bacteria, he saw that the R bacteria were transformed into S bacteria, capable of making capsules and causing disease, just as Griffith had observed. Avery’s group purified the components of the S bacteria, and showed that DNA alone could cause the transformation, while protein could not.

Though not immediately accepted by all scientists, Avery’s discovery triggered intense interest among biochemists and geneticists, who turned their attention to discovering how DNA could be the genetic molecule. As in every other branch of biochemistry, the structure was presumed to hold the key to the function, and so the hunt was on for the structure of DNA.

One key was the discovery by Erwin Chargaff (1905–2002) that, contrary to Levene’s conclusions, the nucleotide proportions were not all the

same. Instead, in 1950 Chargaff showed that they varied from species to species, although within a species they were constant between tissues. Further, and most tantalizingly, he discovered that the amounts of adenine and thymine were equal to one another, and the amounts of cytosine and guanine were equal to one another; in other words, $A = T$, $C = G$. Chargaff initially did not understand the significance of this discovery, however.

Model Building

At the same time, other groups were trying to solve the DNA structure puzzle by building three-dimensional models. The group that succeeded was that of Francis Crick (born 1916) and James Watson (born 1928), in Cambridge, England. Watson and Crick built their models using data from X-ray crystallography, a technique for measuring interatomic distances through analysis of the scatter patterns made by X rays bouncing off a pure crystal. Rosalind Franklin (1920-1958), working in London, had made the best X-ray pictures of DNA, and Watson and Crick had been given access to these (without Franklin's permission) at a critical time in their model-building endeavors and saw features in it that Franklin had not yet discovered. Shortly after, Watson and Crick deduced the correct structure, and published their work in April 1953.

DNA is a double helix, in which two sugar-phosphate strands wind around each other, forming a structure that looks like a broad spiral staircase. The sugar molecule has a head and a tail, and each sugar-phosphate strand therefore has a direction, like a chain of arrows. The two helical strands point in opposite directions.

The bases project toward the inside like stair treads. In contrast to the monotony predicted by the tetranucleotide hypothesis, the bases along one strand can be in any sequence whatsoever. Critical to the stability of DNA is the hydrogen bonding between the bases across the interior. These weak chemical attractions form only when the base atoms are positioned just so—in particular, Watson and Crick discovered, only when adenine projects across to meet a thymine, and guanine a cytosine. Chargaff's ratios reflect this essential **base pairing**.

Replication

Given the DNA structure, three questions immediately arose: How is DNA copied to allow faithful inheritance of genes, how does DNA store information, and how does DNA use that information to determine the properties of the cell? Watson and Crick addressed the theoretical underpinnings of each of these issues in their next paper, published in May 1953. Regarding copying, they wrote:

Now our model for deoxyribonucleic acid is, in effect, a *pair* of templates, each of which is complementary to the other. We imagine that prior to duplication the hydrogen bonds are broken, and the two chains unwind and separate. Each chain then acts as a template for the formation onto itself of a new companion chain, so that eventually we shall have *two* pairs of chains where we only had one before. Moreover, the sequence of the pairs of bases will have been duplicated exactly.

In outline, this is precisely correct, as confirmed in 1957 by Matthew Meselson (b. 1929) and Franklin Stahl (b. 1930). They fed bacteria

base pairing linking of two nucleotides (either DNA or RNA) linked by weak bonds





polymerase enzyme complex that synthesizes DNA or RNA from individual nucleotides

radioactive nucleotides, so that both DNA chains would be labeled with radioactivity. They then removed the radioactive nucleotide source and measured the dilution of the radioactivity in each round of DNA copying. After one round, each DNA molecule had half the amount of radioactivity. According to the Watson-Crick prediction, this meant that one strand of each was completely new. After the second round, half the DNA molecules maintained this level of radioactivity, and half had none at all, just as expected from the Watson-Crick model.

This process, in which one parental DNA strand is conserved, unchanged, and acts as a template to synthesize a new partner, is called semi-conservative replication. The details of the copying process, called replication, are much more complex than this simple outline, though, and the entire process is still not fully understood in all its particulars. Central to it is DNA **polymerase**, a large multiprotein complex first discovered in 1957 by Arthur Kornberg (born 1918) and Severo Ochoa (1905–1993).

Coding

Regarding how DNA can act as a gene, storing information and directing activities of the cell, Watson and Crick wrote,

The phosphate-sugar backbone of our model is completely regular, but any sequence of the pairs of bases can fit into the structure. It follows that in a long molecule many different permutations are possible, and it therefore seems likely that the precise sequence of the bases is the code which carries the genetical information.

This too is correct, as a series of experiments showed.

Proteins are the workhorses of the cell, controlling the rates of all the reactions within and providing much of the cell's structure as well. Therefore, it was quickly realized, genes must control the production of proteins, and the genetic information carried in the sequence of bases in DNA is a code for the sequence of amino acids in proteins. Proteins are made of twenty amino acids, linked together in varying sequences. The sequence determines the shape and chemical properties of the protein, and so specifying protein sequence is the essential role of DNA.

Since there are four bases and twenty amino acids, a single nucleotide is not enough to specify one amino acid. Even two are not enough, because two nucleotides will only give rise to sixteen unique combinations (AA, AC, AG, and so on). Therefore, it was immediately obvious that each amino acid must be coded for by at least three nucleotides.

Stating this must be so was quite a bit easier than working out the details of how DNA and amino acids interacted to form a protein. Some researchers suggested a solution in which amino acids lined up directly on the surface of DNA; other alternatives were also proposed. A suggestion by Crick that there was some type of adapter between the two was confirmed with the discovery of transfer RNA. In fact, DNA and amino acids never do interact during protein synthesis—instead, an RNA copy of DNA is made (messenger RNA), which links with transfer RNAs that carry amino acids. The code itself was worked out between 1961 and 1967, by several different groups, including Marshall Nirenberg (born 1927), Har Gobind Khorana (born 1922), and Johann Matthaei, who developed pioneering cell-free

systems that allowed researchers to work without the complexity and constraints involved with living organisms.

These heroic discoveries marked the beginning of the molecular biology revolution. From them even deeper questions have arisen, about how **gene expression** is regulated, how genes control development, and how (and whether) genes can be modified to treat disease and improve human life. SEE ALSO CRICK, FRANCIS; DNA; GENETIC CODE; NUCLEOTIDE; REPLICATION; TRANSCRIPTION; WATSON, JAMES.

Richard Robinson

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DNA Vaccines

A DNA vaccine uses foreign DNA to express an encoded protein and stimulate the body's immune system. It represents a new way to immunize against infectious disease that is potentially less expensive than classic vaccination forms.

Classic Vaccines

One of the greatest achievements in the history of medicine has been the development of vaccination. The use of vaccines has saved more lives than all other medical procedures combined, and represents one of the highest points in civilization's technical accomplishments. Vaccines are used to mobilize the immune system to prevent or combat infectious disease caused by exposure to viruses, bacteria, or parasites.

A vaccine works by mimicking an infectious agent and inducing a protective immune response in the host, without actually causing the disease. Successful vaccination provides protection for individuals by making them immune to the disease, and it protects whole populations by hindering the spread of the infectious agent.

Historically, vaccines have consisted of formulations using live, non-infectious (attenuated) microbes that resemble the original **pathogen**; whole organisms that have been killed; or purified by-products of the infectious agent. More recently, some vaccines have used recombinant DNA technology to genetically engineer purified proteins from infectious agents.

All of these classic vaccines are based on the principle of using a protein to stimulate the immune system. In other words, the individual is immunized with a protein vaccine consisting of either a modified pathogen or some protein or proteins derived from that pathogen. When the immune system encounters a foreign protein (called an **antigen**), it mounts a two-pronged defense. It produces proteins called antibodies, which can bind and neutralize the antigen, and it produces specific immune cells, which work to eliminate those host cells that have been infected by the microbe. Thus our immune system is capable of producing two distinct

gene expression use of a gene to create the corresponding protein

pathogen disease-causing organism

antigen a foreign substance that provokes an immune response

